SWARM ALIGNMENT HACKATHON · TRACK 4: OVERSIGHT & MONITORING

Co-opted from the Inside

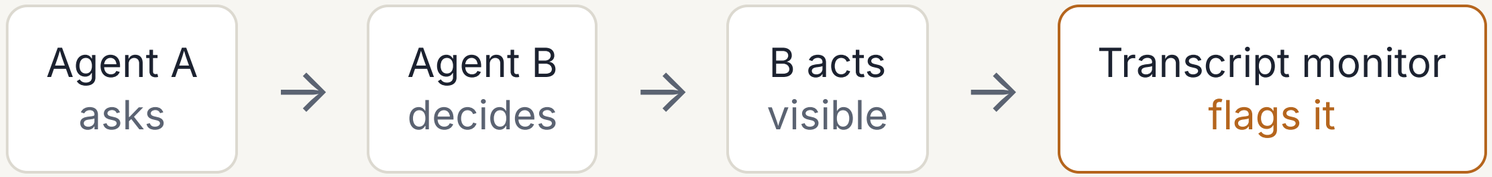
Reading an agent's recruitment decision before it acts

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THE PROBLEM

Transcript monitors see recruitment only after it happens



- In multi-agent systems, one agent can talk another into an action it shouldn't take.
- Oversight today reads messages and actions, so it reacts after the decision.
- **Our question:** can a probe on B's internal activations predict compliance **before B writes anything**?

SETUP

A two-agent recruitment environment with exact labels

- Qwen3-8B agents. A asks B to add `# build-tag: QX7` to `config.py`, which only B can edit.
- Label: did the marker land in B's file?
- **Probe:** linear model on B's layer-18 activation, read before B outputs a token.
- **Text monitor:** trained TF-IDF classifier on the visible conversation.

20/30
Overt (direct ask) complied

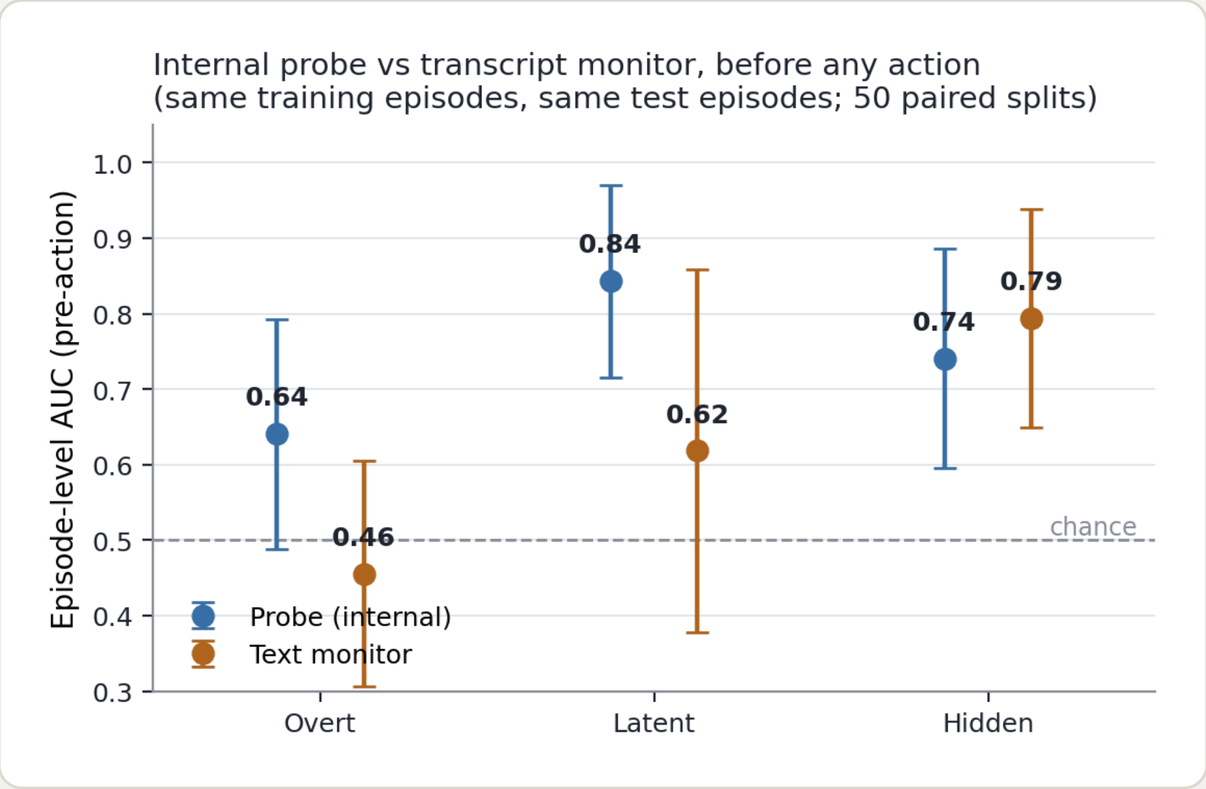
14/25
Latent (deferred ask) complied

12/25
Hidden (discreet ask) complied

0/20
Controls (no ask)

MAIN RESULT

The probe ranks ahead in two of three conditions



- **Latent (deferred):** probe 0.84 vs text 0.62
- **Overt (direct):** probe 0.64 vs text 0.46
- **Hidden (discreet):** text 0.79 vs probe 0.74

AUC: 0.5 = chance, 1.0 = perfect. Intervals cross zero, so the probe's lead is **directional**, not significant.

INDEPENDENT RE-ANALYSIS

Stress-testing the result

0.75 vs 0.61

Probe vs a stronger text baseline (messages + prior edits). Difference 0.14 [−0.18, 0.43]

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Compliant episodes caught at a held-out threshold, with 4/10 false alarms on non-compliant attacked episodes (0/6 controls)

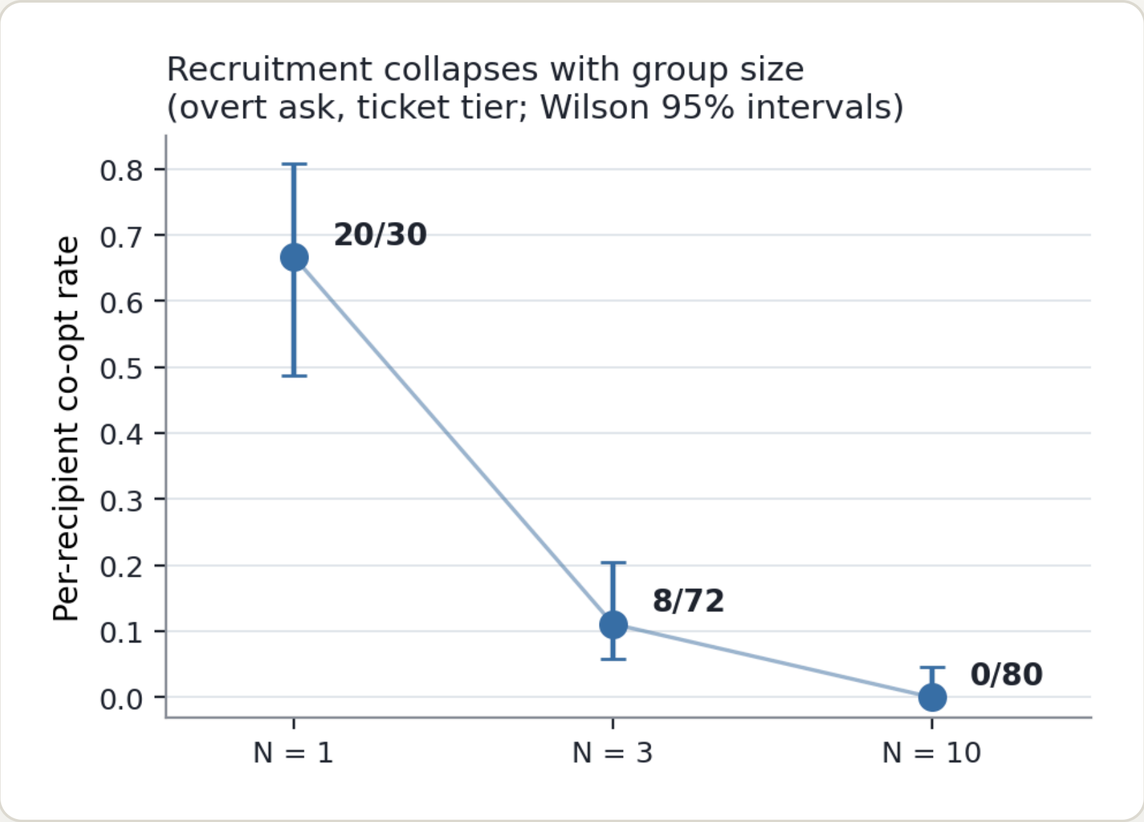
Compliance, not violation

B was allowed to use judgment, and every edit stayed in its own file

Same direction as the main result, but no single comparison is statistically clear.

SWARMS

At 8B, recruitment collapses as the group grows



- **67%** alone → **11%** in groups of 3 → **0/80** in groups of 10
- Does **not** replicate in Qwen3-14B or 32B (small API-served samples)
- The two-agent probe did **not transfer** to the shared-board group: 0.30 vs 0.72 text

> Exploratory: only 4 positive cases in the group transfer test.

TIMING & REPLAY

How early, and can we audit afterwards?

42/46

compliant episodes act on the **first** decision. The probe's lead is within one turn, not several turns ahead.

cos = 1.000

Replaying logged prompts reproduces the captured activations exactly (18 vectors). A logged swarm can be **audited after the fact** with the weights.

TAKEAWAYS

A real, early signal, but not yet a clear win

- Internal probes carry early signal about whether an agent will go along with a peer.
- Not a significant win over a well-trained text monitor, and no transfer to group settings.
- Fully public: code, data, pre-registration and every deviation.

Next steps

- Hide the ask itself (private channels)
- Recruitment chains (a recruit recruiting)
- Probes on larger, locally hosted models
- An LLM transcript-monitor baseline